

Use of adaptive grippers

1. `setGripperState(byte mode, int sp);`

- Function: Set the opening and closing status of the gripper
- Parameter Description:
- mode, jaw opening and closing mode, range 0/1, 0--jaw open to the maximum, 1--jaw closed to the minimum
- sp, jaw opening and closing speed, range 1-100

2. `setGripperValue(int data, int sp);`

- Function: Set the jaw opening and closing angle
- Parameter Description:
- data, jaw opening and closing angle, range 0-100, 0--closed to the minimum angle, 100--open to the maximum angle
- sp, jaw opening and closing speed, range 1-100
- Return value: None

3. `setGripperIni();`

- Function: Set the gripper zero point
- Return value: None

4. `getGripperValue();`

- Function: Get the current angle of the gripper
- Return value: Returns the current gripper angle, ranging from 0 to 100

5. `isGripperMoving();`

- Function: Detect whether the gripper is moving
- Return value: 0: not moving, 1: moving

6. `void setGripperMode(bool mode);`

- Note: This interface is only available for MyCobot320 robots
- Function: Set the adaptive gripper control mode
- Parameter Description:
- mode: mode, 0/1, 0--485 communication control, 1--io control (in io mode, it can only be turned on or off, and the angle cannot be set. When pins 23 and 33 are turned on or off, both pins need to be set to different states, one high and one low)
- Return value: None

7. bool getGripperMode();

- Note: This interface is only available for MyCobot320 robots
- Function: Set the adaptive gripper control mode
- Return value: Adaptive gripper control mode, 0/1, 0--485 communication control, 1--io control

Adaptive robot control

This example will initialize the MyCobotBasic instance, turn on the robot power, and initialize the adaptive gripper. During the initialization process, the gripper will open and close once. After that, the gripper is controlled to reach two specified positions, then open and close once, and the program ends.

```
#include <MyCobotBasic.h>

MyCobotBasic myCobot;

void setup()
{
    myCobot.setup(); // Required API
    delay(100);
    myCobot.powerOn(); // Power on the robot arm
    delay(100);
    GripperInit(); // Initialize the adaptive gripper
}

void loop()
{
    myCobot.setGripperValue(80, 50); // Open the gripper to 80° at a speed of 50
    delay(500); // Once in place, proceed to the next step
    myCobot.setGripperValue(20, 50); // Open the gripper to 20° at a speed of 50
    delay(500);
    myCobot.setGripperState(0, 30); // Fully open the gripper at a speed of 30
    delay(600);
    myCobot.setGripperState(1, 30); // Fully close the gripper at a speed of 30
    delay(600);
}

void GripperInit()
{
    myCobot.setGripperMode(0);
    delay(100);
    myCobot.setGripperState(0, 100); // Fully open the gripper at a speed of 100
    delay(200);
    myCobot.setGripperState(1, 100); // Fully close the gripper at a speed of 100
    delay(200);
}
```

